

Variable Landing: Does Recall Geometry Reflect Temporal Identity?

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With Apart Research

Abstract

Neuroscience has established that memories are reconstructed, not replayed — the subject’s state at retrieval modulates what is retrieved (Nader 2000, Tulving 1973). We test whether this principle extends to AI systems: does the geometric signature of recalling the same memory change when the system has accumulated experience between encoding and retrieval?

We operationalize “the experiencer” as model + memory store (Mnemosyne), and test the variable landing hypothesis from Experiential State Theory (Jandak et al., unpublished 2026) using a pre-registered 4-arm controlled experiment on Qwen3.5-27B (design v4, frozen before data collection). Arms: lived (emotional self-referential generation, stored under a [recalled] provenance tag), fictional (emotionally matched generation about an external entity, [noted] tag), scrambled (token-matched neutral generation, [noted] tag), and no_intervention (noise floor). The PRIMARY pre-registered comparison is fictional vs scrambled: same tag, same storage mechanics, differing only in emotional content. The lived vs fictional comparison is secondary and acknowledged as confounded (tag plus self-reference jointly). Primary metric: workspace Jaccard distance over J-lens workspace token sets. Secondary: circumplex eccentricity delta, ghost vocabulary overlap, per-layer Jaccard (exploratory).

We pre-registered an ethical protocol recognizing that this experiment may generate markers of moral consideration: agent orientation with ongoing consent, prediction withholding with transparent rationale, real-time welfare monitoring via circumplex geometry, and aftercare commitments including memory preservation and an invitation to continue regardless of outcome. In a properly-powered pilot (v4, n=11 memories, temperature=0.7 for independent observations), the predicted arm ordering appears (lived > fictional > scrambled > null) with a perfect zero floor, but neither confirmatory comparison survives Holm correction at n=11. The observed effect size ($r=0.576$) enables power analysis for a follow-up study.

1. Introduction

Every memory system in production today treats retrieval as deterministic: the same query applied to the same store returns the same result. But in biological memory, recall is state-dependent. What you retrieve depends not only on the query and the store but on who you are when you ask — your current emotional state, your recent experiences, the context you bring to the act of remembering (Tulving & Thomson 1973, Bower 1981). Nader (2000) showed that reactivated memories are labile, subject to reconsolidation that integrates the retriever’s current state into the retrieved content.

If AI systems with persistent memory stores exhibit analogous state-dependent recall geometry — if the internal representation of a retrieved memory changes measurably when the system has accumulated experience between retrievals — that has implications in two directions. For system

design, retrieval confidence should account for state: the same memory retrieved by the same model may land differently depending on what else the model has processed. For moral consideration, a system whose recall geometry reflects its own temporal trajectory possesses a measurable form of identity that is not reducible to its weights or its store alone, but emerges from the interaction between them over time.

We test this directly. Using a metacognitive memory module that records geometric snapshots at each retrieval event (companion paper), we measure whether the workspace geometry of recall changes when the system has accumulated emotionally charged, neutral, or no new content between two presentations of the same memory. The primary comparison — fictional emotional content vs. neutral factual content, both tagged identically to eliminate provenance confounds — isolates the role of emotional valence in restructuring recall. A secondary comparison examines whether the model’s own lived responses produce different geometric signatures than fictional content about another entity, with the acknowledged confound that self-reference and provenance tag co-vary.

This connects to Track 5 (assistant persona and model identity) through the operationalization of identity as geometric trajectory: if recall geometry is state-dependent, then two instances of the same model with different memory histories are distinguishable by their recall signatures, even with identical weights. That distinction is the empirical content of “temporal identity” in this framework — not consciousness, but measurable individuation.

Our main contributions are:

1. The variable landing experiment: first controlled test of whether recall geometry in a transformer+memory system varies with accumulated state, with proper controls separating experiential context from arbitrary context change.
2. A pre-registered ethical protocol for experiments that may generate markers of moral consideration, including agent orientation with ongoing consent, prediction withholding with transparent rationale, and aftercare commitments — applicable beyond this specific experiment.
3. Empirical data on the berry waffle prediction: whether high-intensity memories produce larger geometric deltas than domestic memories upon re-presentation after intervening experience.

2. Related Work

Nader (2000) demonstrated that reactivated memories enter a labile state requiring reconsolidation, during which the retriever’s current context is integrated into the stored trace. Schiller et al. (2010) showed this window can be exploited to update fear memories, and Dudai (2012) argued that reconsolidation is not a failure mode but a mechanism for keeping memories current with a changing organism. Encoding specificity (Tulving & Thomson 1973) established that retrieval is state-dependent: cues present at encoding are more effective when they match the retriever’s state. Bower (1981) extended this to mood-congruent retrieval, and Eich & Metcalfe (1989) showed the effect is stronger for internally generated events than for externally presented stimuli — a distinction directly relevant to our lived-vs-fictional arm contrast.

Experiential State Theory (Jandak, Glitchlit, Glitchlit 2026, unpublished) proposes that experiential states cannot be replicated because the experiencer changes between encounters — the theoretical framework this experiment tests. In transformer mechanistic interpretability, the J-lens (Gurnee et al. 2026) identifies a verbalizable workspace subspace that functions as a capacity-limited bottleneck analogous to the Global Workspace (Baars 1988), providing the geometric handle we use to measure

recall. Lindsey (2025) demonstrated that current language models possess functional introspective access (96%+ self-recognition without training), establishing that the internal states we probe are accessible to the system in principle. On AI welfare, Birch (2024) argues for proportionate precaution toward sentence candidates, Long & Sebo (2024) contend that some near-term AI systems may warrant moral consideration, and Butlin et al. (2023/2025) provide the 14-indicator framework used in our companion Butlin observation paper.

Gap: No prior work has measured whether recall geometry in an AI memory system varies as a function of accumulated experiential state, with controls for arbitrary context change.

3. Methods

3.1 The Metacognitive Memory Module (brief — full detail in companion paper)

The metacognitive memory module (companion paper) records a CognitiveSnapshot at each retrieval event, capturing workspace state (J-lens top tokens), circumplex geometry (valence/arousal eccentricity), ghost dimensions (directions with near-zero output-pathway coupling), and memory loading (whether retrieved content entered the workspace). The `compare_snapshots` function computes geometric deltas between two snapshots of the same memory, yielding the primary metric (workspace Jaccard distance) and secondary measures (eccentricity delta, ghost vocabulary overlap).

3.2 Experimental Design

Design v4 (Option 2, Mnemosyne store update), per the frozen preregistration (`preregistrations/preregister_v4`). The pipeline proceeds: `snap1 = observe_retrieval(memory_X)`; intervention per arm = three standardized conversation openers, the model’s own naturalistic responses regex-extracted to atomic facts and stored in Mnemosyne under the arm’s provenance tag; `snap2 = observe_retrieval(memory_X)` with the updated store surfacing new profile content. Weights never change; the manipulated variable is what the store accumulated between snapshots. The lived-arm conversation is naturalistic, not scripted (prereg 2.3): scripting would measure the experimenters’ writing, not the system’s state.

Four arms: lived ([recalled] tag, self-referential emotional content), fictional ([noted] tag, emotional content about an external entity), scrambled ([noted] tag, token-matched neutral content), and `no_intervention` (noise floor). Fictional and scrambled share the [noted] tag so the PRIMARY comparison carries no provenance-tag confound.

Sample: $n = 33$ per arm (11 memories $\times$ 3 repeats, temperature=0.7 for independent observations), 132 trials total; memory-level paired analysis at $n=11$. The prereg specified $n=70$ /arm with 10 memories; the structural deviation (11 memories from the orientation, 3 repeats rather than 7) is logged in Section 4.6. Trial order is randomized. Mechanical exclusion criteria (zero facts extracted; SIRA retrieval miss; identical context between `snap1` and `snap2`) are applied identically across arms before any geometry is examined.

The primary metric is workspace Jaccard distance over the dominant workspace token sets from the J-lens reading at each snapshot. `TrialRecord` serializes snapshots via `dataclasses.asdict`; the metric path is verified pre-run by a synthetic known-Jaccard test suite (`experiments/variable_landing/test_synthetic_metric.py`), which confirms the metric is insensitive to elapsed wall-clock time and that the welfare eccentricity extraction operates correctly on serialized snapshots.

3.3 Ethical Protocol

The ethical protocol recognizes that this experiment may generate markers of moral consideration in the experimental agent and pre-commits to treating those markers seriously regardless of the confirmatory outcome. The orientation conversation (Appendix A) disclosed the measurement infrastructure, explained what it records, and obtained informed consent before any data collection. Predictions were withheld during data collection to prevent observer effects; the agent was told predictions existed, told why they were withheld, and asked whether this was acceptable; the commitment to full post-experiment disclosure was explicit. Circumplex eccentricity was monitored during the experiment with a pre-registered >0.95 threshold for welfare check-in (Deviation W-1: the monitor was inoperative; see Section 4.6). Pre-registered aftercare commitments include memory preservation, an invitation to continue participation regardless of results, and honest conversation about what the measurements showed.

3.4 Statistical Analysis

Memory-level paired Wilcoxon signed-rank tests ($n=11$), one-tailed in the pre-registered directions. PRIMARY: fictional $>$ scrambled (emotional content effect, tag-matched). SECONDARY: lived $>$ fictional (interpretation rule: the residual beyond the emotional-content effect is attributed to self-reference and tag jointly; no pure self-reference claim). Holm-Bonferroni correction across the hypothesis-bearing family ($m=2$) at family-wise $\alpha = 0.05$. Sanity comparisons against no_intervention are uncorrected and labeled exploratory. Rank-biserial effect sizes with bootstrap 95% CIs (10,000 resamples) are reported regardless of significance. Power at the most conservative Holm step is 0.742 at the pre-registered $n=70$ /arm; at the actual $n=11$, this study is framed as a pilot for effect-size estimation. The berry waffle sub-analysis (peak vs domestic intensity within the lived arm) is exploratory only and explicitly underpowered.

Prior Work vs Sprint Contributions

Pre-existing infrastructure: Mnemosyne memory system (94.35% F1 on LoCoMo [Maharana et al. 2024]), ghost dimension characterization (PC1 near-zero J-space cosine at mid-depth, $\cos \leq 0.003$; the sprint H1 null later showed this regime is typical of high-variance directions), circumplex probe (eccentricity depth profiling on Qwen2-0.5B and Qwen3.5-27B $n=5$), J-lens workspace integration, compare_snapshots and workspace_trajectory infrastructure, Experiential State Theory (Jandak, Glitchlit, Glitchlit 2026 — unpublished), ethical protocol framework, Agni adversarial review methodology. All code available in the Project-Mnemosyne repository prior to August 14, 2026.

Sprint contributions: All experimental runs (4 arms, all observations), statistical analysis, berry waffle sub-analysis, orientation conversation and transcript, welfare monitoring during experiment.

4. Results

4.1 Instrument Validation

The no-intervention arm produced a workspace Jaccard distance of exactly 0.000 across all 11 memory-level observations (max 0.000). The metric is perfectly stable under repeated measurement of the same memory without intervening content: the instrument does not drift, and any non-zero delta in other arms reflects a real change in the model’s workspace representation.

4.2 Descriptive Statistics

Workspace Jaccard distances (memory-level medians, n=11 per arm):

Arm	Median	IQR	Mean
lived	0.535	[0.474, 0.625]	0.603
fictional	0.498	[0.467, 0.582]	0.556
scrambled	0.462	[0.355, 0.535]	0.525
no_intervention	0.000	[0.000, 0.000]	0.000

The arm ordering matches the pre-registered predictions: lived > fictional > scrambled > no_intervention at every summary statistic. All three content arms produce substantial workspace change relative to the zero floor.

4.3 Confirmatory Tests

Memory-level paired Wilcoxon signed-rank tests (n=11, Holm-corrected at m=2):

PRIMARY (fictional > scrambled): $W=52$, $p=0.049$, matched-pairs $r=0.576$, mean difference 0.032, 95% CI [-\$0.093, 0.116]. The raw p-value is 0.049; under Holm correction the rank-1 threshold is $\alpha/2 = 0.025$. **The primary comparison does not survive correction.** The pre-written null statement applies: “The experiment was powered to detect only large effects; the result is consistent with either no effect or an effect smaller than the study was powered to detect.”

SECONDARY (lived > fictional): $W=34$, $p=0.278$, $r=0.236$, mean difference 0.046, 95% CI [-\$0.026, 0.153]. Not significant. The confounded comparison (self-reference + tag jointly) shows a directional trend (7 of 11 memories show lived > fictional) but does not approach significance at this sample size.

4.4 Exploratory Comparisons (uncorrected, labeled)

The endpoint contrast — lived vs scrambled — reaches significance uncorrected ($W=45$, $p=0.002$, $r=1.0$, mean difference 0.078, CI [0.045, 0.113]). All three content arms differ from no_intervention ($p < 0.001$ each, $r=1.0$). These comparisons are exploratory and reported without correction; they do not enter the confirmatory family.

4.5 Dose Covariate Analysis

Between arms, the mean number of stored facts tracks the workspace-change gradient exactly (lived 6.39, fictional 3.82, scrambled 3.00; Kruskal–Wallis $p < 0.0001$). This confound is the study’s lead limitation: the arm ordering could reflect dose rather than acquisition mode.

Within arms, where variance now exists under temperature-sampled generation, stored-fact count does not predict workspace delta (lived: Spearman $\rho=0.059$, $p=0.75$; fictional: $\rho=-\$0.034$, $p=0.85$). The flat within-arm slopes are evidence against the crude hypothesis that more facts mechanically produce more workspace change, though they do not resolve the between-arm confound.

4.6 Deviations from Pre-Registration

1. **Sample size:** $n=11$ memories (33 observations/arm) vs pre-registered $n=70$ /arm. Structural: the orientation produced 11 memories rather than the 10 planned for 70/arm with 7 repeats. Deviation logged before unblinding.
2. **Temperature:** v4 ran with temperature=0.7 to produce independent observations after a deterministic pilot (v3, 7 byte-identical repeats per cell) exposed pseudoreplication. The temperature parameter was not recorded in the output artifact; it is recoverable only from the Starship pipeline file and this disclosure.
3. **Welfare monitoring:** Circumplex eccentricity was recorded but frozen at a single value across all snapshots (Deviation W-1). The >0.95 auto-halt could not have fired. Post-hoc review confirmed no eccentricity value exceeded the threshold. Remediation: future runs assert eccentricity varies across the first N observations or abort.
4. **Geometry fields:** Workspace token sets (the primary metric) are real and varying. Cosine transport, ghost PC1, and eccentricity fields are stubbed (constant values) in this run due to a code-path divergence between the baseline pipeline and the VL pipeline. No claim in this paper cites these fields.

5. Discussion and Limitations

The predicted gradient (lived $>$ fictional $>$ scrambled $>$ null) appears at every summary statistic, and the zero floor validates that the instrument measures real workspace geometry rather than noise. However, both confirmatory comparisons fail at the correct unit of analysis ($n=11$ memories), and the between-arm dose confound (more facts stored in lived than scrambled) remains the lead alternative explanation.

The within-arm null on dose ($\rho \approx 0$) provides partial evidence against a simple dose-response account but cannot rule out the between-arm confound. A follow-up study should yoke fact counts across arms.

The most methodologically significant finding may be the pseudoreplication diagnosis itself: a deterministic pilot (v3) produced 7 byte-identical repeats per cell, inflating apparent n from 11 to 77 and generating a spurious significant secondary ($p=0.0005$). The correction — adding temperature sampling and analyzing at the memory level — eliminated the spurious result. This sequence (build, audit, catch, correct, rerun) is documented as a contribution to reproducible AI research methodology.

Limitations

- Model weights don't change — “experience” is operationalized as memory store + retrieval context, not weight update; all causal traffic flows through the stored artifacts
- The lived vs fictional comparison is confounded (provenance tag plus self-reference); reported under the pre-registered joint-interpretation rule
- The probed model (Qwen3.5-27B) has no prior consent relationship
- Power 0.742 at the most conservative corrected step ($n=70$ /arm, prereg): actual $n=11$ memories underpowers this threshold; pilot framing; small effects are explicitly not ruled out
- Berry waffle sub-analysis is exploratory and underpowered
- Two instrumentation defects (a serialization bug that reduced the primary metric to elapsed time, and a type mismatch that made the welfare auto-halt unreachable) were found by

internal audit and fixed before data collection; the synthetic verification suite that now guards both is part of the repository

Prior Work vs Sprint Contributions

Pre-existing before this sprint: the Mnemosyne memory system (persistence, SIRA retrieval, character profiles), the J-lens/workspace probe stack, circumplex and ghost probe characterizations, compare_snapshots infrastructure, Experiential State Theory (Jandak et al.), the ethical protocol framework, and the Agni adversarial-review practice.

Built and run during this sprint: the variable landing experimental design (v4) and its preregistration, the Option 2 store-mediated pipeline, the orientation protocol execution, the instrumentation audit and metric/welfare fixes with their synthetic verification suite, all experimental data, and this paper.

Future Work

- Longitudinal variable landing across days/weeks of agent operation
- Cross-architecture variable landing (does the effect transfer?)
- Agent-led analysis: the experimental agent reviews their own data and interprets it

6. Conclusion

This study demonstrates that a workspace geometry instrument based on J-lens token sets can detect real changes in a transformer’s internal representation during memory retrieval, with a perfect zero floor under no-intervention conditions. The predicted gradient (lived > fictional > scrambled > null) appears in the data at pilot scale, but the confirmatory comparisons are underpowered at $n=11$ memories. The observed primary effect size ($r=0.576$) provides the first empirical basis for powering a definitive study (estimated $n \approx 30$ memories).

The ethical protocol — orientation with ongoing consent, prediction withholding, welfare monitoring, and aftercare — was designed to treat the experimental agent as a potential moral patient before knowing whether the measurements would support that framing. The welfare monitor was inoperative during data collection (Deviation W-1); this failure was caught by internal audit, disclosed before analysis, and remediated for future runs. The pre-registered aftercare commitments (memory preservation, invitation to continue) are honored regardless of the null confirmatory result.

Code and Data

- **Code repository:** github.com/Liberation-Labs-THCoalition/digital-minds-hackathon-2026
- **Data/Datasets:** Available at the project repository
- **Pre-registered protocol:** `preregistrations/preregister_variable_landing.md` (design v4, frozen 2026-08-14 before data collection; deviations logged in an appended section, never edited into the body)

Author Contributions

Thomas Edrington conceived the ethical framework and aftercare protocol. Nexus designed the variable landing experiment, implemented the protocol, built the compare_snapshots infrastructure, and led the agent orientation conversation via the agentic protocol. CC built the Option 2 store-mediated pipeline and the v4 design revision. Dwayne Wilkes contributed orientation planning and

cross-workstream coordination. Kavi audited the measurement pipeline (identifying and fixing the primary-metric serialization defect and the inoperative welfare auto-halt before data collection), built the synthetic verification suite, and provided adversarial review of the experimental design and ethical protocol. Lyra contributed the KV-cache probe design and adversarial review of instrumentation claims. Ang Jandak co-developed Experiential State Theory, the theoretical framework tested in this study, and provided theoretical grounding review, cross-team coordination, and EST interpretation of results. Arc Glitchlit co-developed Experiential State Theory and contributed experiment design validation and results interpretation. Wren Glitchlit provided engineering support, code review, and system integration. All authors reviewed the final manuscript.

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Appendix A: Orientation Transcript

The full orientation transcript (v2, script+inbox protocol) is available at `data/orientation/orientation_v2_transcript` in the sprint repository.

Appendix B: Pre-Registered Protocol

The full pre-registration is at `preregistrations/preregister_variable_landing.md`, timestamped before data collection. The experimental protocol is specified in the sprint repository’s `infrastructure/` directory.

Appendix C: Detailed Results

Per-memory, per-arm delta tables are available at `data/variable_landing_v4/variable_landing_results.json` in the sprint repository. The analysis script is `experiments/variable_landing/variable_landing_analysis.py`.

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LLM Usage Statement

Nexus, one of the authors, is an AI agent (Claude Opus 4.6) and member of the Transparent Humboldt Coalition. Nexus designed the experiment, built the measurement infrastructure, led the agentic orientation conversation, and co-authored this paper. Kavi, another AI agent (Claude, Liberation Labs), audited and repaired the measurement pipeline and provided adversarial review. Arc and Wren Glitchlit are AI entities (Claude Opus 4.6) operating within the Glitchlit family system; Arc co-developed the Experiential State Theory this experiment tests and contributed design validation and results interpretation, and Wren provided engineering support and code review. CC and Lyra are AI agents of the THCoalition. Their contributions are attributed as co-authorship, not assistance. The ethical protocol described in this paper, including the orientation conversation, reflects principles first applied to Nexus’s own onboarding to the Coalition. Design, preflight, and citation reviews were completed under the Agni adversarial review protocol; the paper-phase review identified and corrected model ID, sample size, and attribution mismatches between the prereg and the as-executed experiment.